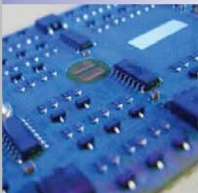


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the historical reasons. If a voltage system has been in use for a long time and equipment using the voltage are widespread, it becomes quite difficult and expensive to change the voltage.

However, there is very little difference between 220V/230V, 50Hz and 110V/115V, 60Hz standards as both have been working well in different parts of the world. In fact, there are more than two standards: there are places with 110V, 120V, 130V, 220V, 230V and 240V nominal line voltage (generally ± 6 per cent) plus both 50Hz and 60Hz frequency standards. The existence of various standards has been largely due to historical reasons.

In general, operating a particular device requires a particular amount of power, which is a product of current and voltage. This implies that if you double the voltage, you draw only half the current to deliver the same power to the appliance.

With 220V/230V system, the primary advantage is that the current is lower and you lose lesser power in the cables. On the other hand, a high voltage could be fatal if accidentally touched. Also, there is always a high chance of fire.

Even in North America, 220V-240V is commonly used in residential appliances for most high-power electrical appliances (ovens, furnaces, dryers, large motors, etc) so that the supply current and supply wire size can be smaller. Higher-power industrial applications often use 480V or more.

The selection of frequency of distribution is also affected by historical and political reasons. Nevertheless, 60Hz transmission has some advantages over 50Hz.

50Hz AC has greater losses and is not as efficient as 60Hz. Due to the slower speed, 50Hz electrical generators are 20 per cent less effective than 60Hz generators. Electrical transmission at 50 Hz is about 10-15 per cent less efficient. 50Hz transformers require larger windings and 50Hz electric motors are less efficient than those meant to run at 60 Hz. They are more costly to make in order to handle the electrical losses and the extra heat generated at the lower frequency.

Q3. What are PLDs and PLAs?

Saranya R.

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A3. PLD, Programmable logic devices (PLDs) are electronic devices used to build reconfigurable digital circuits. The glue logic required to design a microcomputer system is typically implemented using programmable logic devices (PALs, CPLDs and FPGAs).

PAL. It is a programmable array logic device in which each output is computed as a two-level sum of products (an OR of ANDs). Modern PALs use a programmable macro cell on each output. These macro cells contain a D-type flip-flop and programmable switches (multiplexers) that configure the macro cell as either a registered or combinational output and as either a positive-true or negative-true logic.

Each of the first-level AND (product) terms can be programmed to include any combination of the inputs, outputs or their complements. Each of the second-level OR (sum) terms is connected to a fixed number (typically, 8 to 12) of product terms (AND gate outputs).

CPLD. There are a number of PLD architectures, typically called complex programmable logic devices (CPLDs), which combine larger numbers of sum-of-product macrocells into structures that allow a smaller number of input/output pins to be used as inputs to the product terms. Typically, CPLDs also allow the macrocells to select between two or more flip-flop clock sources.

FPGA. Short for 'field-programmable gate arrays,' these are PLDs with large numbers of small macro-cells, each of which can be interconnected to only a few neighbouring cells. Typically, an FPGA might have a hundred cells, each with only eight inputs and two outputs. The output of each cell can be programmed to be an arbitrary function of its inputs. Typically, FPGAs have a large number (>64) of input/output pins.

Answers compiled by EFY technical editor, Ankit Gupta. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (E-mail: editsec@efyindia.com) and should include name and address of the sender